Reference Event 1: The Destruction of the Kakhovka Dam, June 2023

Characterization Against the Dams Sector Design Basis

DBA-DAMS-RE-1 — Reference Event Instrument, DBA-DAMS Series

Lineage: Issued under the Dams Sector Framework v0.4; DBA-MA Parent Framework; conferred on the proposal of Dam Structures and Water-Control Works (DBA-DAMS-DS-FC1)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the destruction of the Kakhovka dam on the Dnipro on 6 June 2023 back through DBA-DAMS-DS-FC1 (*Dam Structures and Water-Control Works*), whose governing military-action scenario it characterizes (§6.3). It asks whether the class's design basis anticipated what happened, and returns what it finds to the class and to the Dams Sector Framework.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Dams Sector Framework v0.4, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 12 October 2026, on the proposal of *Dam Structures and Water-Control Works*. It is the series' first reference event instrument, and the first written directly in Project 11 rather than migrated. Section references of the bare form §N are to *Dam Structures and Water-Control Works* v0.4, the version this instrument read.

The scope runs from the taking of the dam by Russian forces in the opening days of the full-scale invasion of February 2022, through the anomalous operation of the dam from November 2022 and the breach of 6 June 2023, to the joint needs assessment of 17 October 2023 and the restoration of a cooling-water supply at the Zaporizhzhia nuclear plant by September 2023. The war continues, the count of the dead is contested and no finding of any court bears on the event, so the record is declared open at the record date of 13 October 2026. Times are local (UTC+3).

The attribution posture is fixed. The Government of Ukraine stated that Russian forces destroyed the dam; the Government of the Russian Federation stated that the destruction was deliberate sabotage by the Ukrainian side. Each statement is its government's own, and this instrument makes no finding of attribution and none of fault. No individual is named. Every count is attributed to whoever published it. Where the sources differ, Section 2.2 states each position as its author's.

The structure, its spillway and gates, and the release of its reservoir lie within the dam-structures class (§1.2, §8.3). The destroyed powerhouse and the loss of its generation are *Large Hydroelectric Storage*'s (DBA-ES-HY-FC1); the water drawn from the reservoir is *Storage Reservoir*'s (DBA-W-FC3); the cooling water of the Zaporizhzhia plant is the nuclear series of DBA-MA's. The constructs applied are defined in the Foundational Definitions and cited, not restated: FD-BL (*The Bright Line*), FD-BR (*The Broker*), FD-LD (*Layer Decomposition and Determinism*), FD-LS (*Local Sovereignty*), FD-BL-D1 (*Mode of Discharge*) and FD-BL-D2 (*Indication Integrity*). FD-REG holds their versions.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary sources are the joint Post-Disaster Needs Assessment of the Government of Ukraine and the United Nations, released on 17 October 2023; the International Atomic Energy Agency's Updates 161 and 162 of 6 and 7 June 2023 and its later statements on the Zaporizhzhia plant's water supply; the statement of the Secretary-General of the United Nations of 6 June 2023; the statement of the United Nations Humanitarian Coordinator for Ukraine of 18 June 2023; NORSAR's statement on the seismic signals; and Yang and colleagues, *Pre-failure operational anomalies of the Kakhovka Dam revealed by satellite data*, Communications Earth & Environment 5, article 230, 2 May 2024, a peer-reviewed reading of satellite altimetry and imagery. NORSAR's own page could not be reached in this session, and its statement is read as the press carried it. Governments' statements are read as their own. The Associated Press investigation of 28 December 2023 and NPR's report of 10 February 2023 are cited where used. No non-public material is used.

2.2 Where the Sources Differ

On the cause, the two governments' statements are opposed (Section 1.2). NORSAR reported that its regional stations recorded signals of an explosion at 02:54, whose time and location coincided with reports of the dam's collapse, with a weaker signal at 02:35; it did not specify what caused the explosion, and its scientists were reported as uncertain whether the explosion caused the collapse. Yang and colleagues found that the dam had been operated anomalously from November 2022, that its spillway had been overtopped continuously from late April 2023, and that the structures downstream of the spillway showed signs of being compromised before the breach; they did not determine what breached it. This instrument does not reconcile the explosion signal with the progressive damage. It reads both, because each is a path the design basis must cover (Finding F-3).

On the dead, the counts differ by an order of magnitude. Ukraine's Minister of Internal Affairs gave 31 killed in territory controlled by the Government of Ukraine, 29 in Kherson oblast and two in Mykolaiv, as reported on 20 July 2023. The Russian Federation stated that 59 people drowned in the territory it controls. The Associated Press concluded on 28 December 2023 that the dead in the occupied town of Oleshky alone were at least in the hundreds, local health workers estimating 200 to 300, and that the occupation authorities had undercounted them and taken control of the death certificates. Each count is its publisher's; this instrument adopts none, and no finding rests on an exact number (Finding F-7).

3. The Event

3.1 The Dam and Its Reservoir

The Kakhovka hydroelectric plant's dam was the lowest of the Dnipro cascade, holding one of the largest reservoirs in Europe. Above it on the reservoir's shore stood the Zaporizhzhia nuclear plant, which drew its cooling water from the reservoir; below it lay the city of Kherson and the low ground of both banks to the Black Sea. The reservoir also fed water systems upstream and downstream and the North Crimean Canal. The dam is of the class's first reference structure: a concrete dam with a hydroelectric plant and a gated spillway (§1.3).

3.2 The Holding of the Dam

Russian forces took the dam in the opening days of the full-scale invasion in February 2022 and held it until its destruction, some fifteen months. Its owner, Ukrhydroenergo, could neither operate nor inspect it. On 11 November 2022, as Russian forces withdrew from the right bank, a segment of the road carried over the dam was destroyed (NPR; Yang and colleagues). From then the reservoir was managed by whoever held the gates on the left bank.

3.3 The Pool Under the Holder's Operation

Yang and colleagues found that from November 2022, for more than six months, the spillway's activation stayed at about 15 percent of its capacity, through the same gates, where in normal years the gates are opened to 30 to 80 percent for less than two months in early summer. The reservoir first fell, reaching in late January 2023 its lowest level in three decades of satellite observation; NPR, on the same altimetry, put it at about 14 meters on 10 February 2023, some two meters below normal, with water pouring through gates on the held side, and reported the regional authority's statement that the plant's cooling would be at risk below 13.2 meters. The reservoir then rose, reaching in April 2023 its highest level in three decades. On the authors' weir-flow model, the outflow exceeded the spillway's design capacity by 22 April, and the spillway was overtopped from about 23 April until the breach. The other reservoirs of the cascade stayed within their usual ranges.

3.4 The Damage Before the Breach

On satellite imagery, Yang and colleagues observed damage to a pier downstream of the gates on 28 May 2023; signs on 2 June that the left-bank bridge and its supporting structures were compromised; and further parts missing on 4 June. They attribute the pattern, as likely, to erosion of the spillway's foundation by the prolonged overflow, and record that no countermeasure or hazard warning was issued.

3.5 The Breach

In the early hours of 6 June 2023 the dam was breached and its reservoir began to drain. NORSAR's statement of the signals is at Section 2.2. The President of Ukraine said that an internal explosion had occurred at 02:50. The Secretary-General of the United Nations called the destruction "another devastating consequence of the Russian invasion of Ukraine" and "a monumental humanitarian, economic and ecological catastrophe", and said that the United Nations had no access to independent information on its circumstances.

3.6 The Flood

The joint needs assessment counts about 620 square kilometers submerged across four oblasts, Kherson, Mykolaiv, Dnipropetrovsk and Zaporizhzhia, and about 100,000 residents directly affected, with damage and losses of nearly 14 billion United States dollars and recovery needs of about 5 billion. The Russian Federation stated that about 408 square kilometers of the flooded area lay in the territory it controls (Associated Press), about two-thirds of the assessment's figure.

3.7 The Warning, the Evacuation and the Response

Evacuation from the right bank was organized by the Ukrainian authorities. On the left bank, the Associated Press reports that the occupation administration first called the situation "not critical", that its services were absent for the first days, and that residents who were offered evacuation into Russia and asked to be moved within the town were refused. Evacuations came under shelling: the President of Ukraine called the Russian shelling of evacuation points in Kherson "absolutely deliberate", and the authorities installed by Moscow on the left bank said that two evacuees were killed by Ukrainian shelling (Moscow Times, 9 June 2023). On 18 June the United Nations Humanitarian Coordinator for Ukraine stated that the Russian Federation had declined the United Nations' requests for access to the areas under its control and urged it to meet its obligations under international humanitarian law; the Kremlin's spokesman answered that shelling made a visit too dangerous.

3.8 The Loss of the Pool

The reservoir fell fast. The International Atomic Energy Agency reported on 6 June that the level at the Zaporizhzhia plant had fallen 83 centimeters to 15.44 meters between 10:00 and 20:00, and on 7 June that it had fallen about 2.8 meters to 14.03 meters, the rate having peaked at about 11 centimeters an hour. Below 12.7 meters the plant could no longer pump from the reservoir. The Agency reported that the plant's large cooling pond was full and could supply it for several months, its six reactors being in shutdown, that the plant was refilling its sprinkler ponds and channels while the reservoir could still be reached, and that the pond's integrity had to be maintained; its Director General saw no short-term risk to nuclear safety. By September 2023 ten groundwater wells had been drilled at the site, with an eleventh planned, to supply the sprinkler

ponds, the cooling pond still intact (International Atomic Energy Agency, as NucNet reported on 25 September 2023). The United Nations' Under-Secretary-General for Humanitarian Affairs put the people lacking access to drinking water at about 700,000, as reported on 10 and 11 June 2023. The hydroelectric plant was destroyed with the dam.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-5) and effect elements (E-1 to E-7), as Table 4-1 sets out. Composed, they are the class's governing military-action scenario (§6.3), the deliberate breach of a high-hazard concrete dam at a high pool, opening Chain I and Chain P from one breach (§3.2), preceded by fifteen months in which the structure was held, its pool set and its inspection denied (§6.2).

Table 4-1. Event elements and their mapping. Section references are to *Dam Structures and Water-Control Works* v0.4.

No.	What happened	Class provision	Section
I-1	The dam taken in February 2022 and held for some fifteen months; the owner can neither operate nor inspect it	Persistence; denial of inspection	3.2
I-2	Road segment over the dam destroyed on 11 November 2022; the gates thereafter worked from the held bank	Persistence; works under the holder's control	3.2
I-3	Spillway held at about 15 percent; pool drawn to a three-decade low in January 2023, then raised to a three-decade high in April	Governing release at the pool the event assumes (§3.3, §4.2)	3.3
I-4	Spillway overtopped from about 23 April; damage downstream of the gates visible from 28 May; no warning issued	Persistence through denial of inspection (§6.2); §5.3	3.3, 3.4
I-5	Breach on 6 June; signals NORSAR reported as an explosion at 02:54	Deliberate breach (§6.3)	3.5
E-1	About 620 square kilometers submerged and about 100,000 residents directly affected	Chain I; inundation footprint (§3.3, §4)	3.6
E-2	The dead counted by three publishers, from 31 to at least hundreds	Exposure (§4.5)	2.2
E-3	Evacuation on one bank; absence and refusal on the other; evacuations shelled; humanitarian access declined	Denial (§6.2); evacuation for the deliberate-case footprint (§6.3)	3.7
E-4	Reservoir falling up to about 11 centimeters an hour toward the 12.7-meter pumping limit	Chain P (§3.2)	3.8
E-5	Nuclear plant cooled from its own pond, later supplemented by wells	Pool-dependency margin (§3.3, §6.3); §6.6	3.8
E-6	About 700,000 people without access to drinking water	Chain P; dependents (§8.1)	3.8
E-7	Hydroelectric plant destroyed with the dam	Boundary with *Large*	3.8

		Hydroelectric Storage (§8.3)	

5. Tier Determination

The four characteristics are read as §6.2 states them for the class.

5.1 Intentionality

Present, on the record, without attribution. Both governments stated that the dam was destroyed deliberately, each naming the other, and NORSAR reported the signal of an explosion at the time of the breach. This instrument does not determine who acted, and does not determine whether the explosion or the progressive damage of Section 3.4 breached the structure (Finding F-3).

5.2 Persistence

Present, in both of the forms §6.2 states. The structure was held for some fifteen months, and through that holding its owner was denied its maintenance and inspection; the works were operated by the holder for seven months in a way that first emptied and then overfilled the reservoir (Findings F-1 and F-2).

5.3 Precision

Not determined. The breach came at a three-decade high pool, which §3.3 and §4.2 take as the adversary's choice; the record does not show whether the moment or the pool was chosen, and this instrument does not infer it.

5.4 Denial

Present. The downstream response on one bank was absent for days and access for the United Nations was declined; evacuations on both banks came under shelling, each side's officials attributing it to the other (Section 3.7).

5.5 Determination

The event does not carry the military-action tier. Under the author's rulings of 10 and 13 October 2026, recorded at FD-DBL (*Design Basis Layers*) §6.4, all four characteristics must be established on the record, and a characteristic the record cannot determine counts as not present. Intentionality, persistence and denial are present in the forms §6.2 states, and precision is not determined; the structure was held throughout by a party to an armed conflict. The event is recorded as military action realizing the class's governing scenario (§6.3), the deliberate breach at high pool, in both chains, at the reach of four oblasts and of a nuclear plant's cooling water, with the attribution each government's own. It is the scenario, not the tier, that the record supports; if later evidence establishes precision, this instrument is revised and the tier is given

then. It is the series' first reference event and its first military-action event. That is the tier determination returned for the DBA Reference Event Register. The impacted sectors proposed for the register's column are Energy, for the hydroelectric plant (DBA-ES-HY-FC1); the nuclear series of DBA-MA, for the Zaporizhzhia plant's cooling water; and Water, for the supply drawn from the reservoir.

6. Characterization Against the Design Basis

6.1 §2.4, §3.3, §6.2 and §6.3 The Class's Account of the Event

Held, with a gap. The account is accurate in direction and in every figure checked. Four points are qualified and go to the class as Erratum 1 of Project 11's note of 13 October 2026 (Dams batch 1): NORSAR's reading, the pool's level, the United Nations' access and the plant's water supply.

6.2 §2.4 and §3.2 The Two Chains from One Breach

Held. The flood went down the valley and the loss of the pool went up the reservoir, from one breach, as the class reads Kakhovka.

6.3 §3.3 and §4.2 The Governing Release at the Highest Permitted Pool

Held, with a gap. The breach came at a three-decade high, as the class's deliberate case assumes; but the pool had been set there over months by the party holding the gates, above what the spillway could pass, and not by any permission. Finding F-1.

6.4 §6.2 Persistence Through a Long Denial of Inspection

Held, with a gap. The v0.4 sentence fits the event: fifteen months held, the owner unable to inspect, the structure degrading. The event adds that the holder's operation of the works is itself a form of persistence, and that a denied structure was still observed, from orbit, by people with no duty to warn. Finding F-2.

6.5 §6.1 and §6.3 The Deliberate Breach as the Governing Scenario

Held, with a gap. The scenario is realized on any reading of the cause, and the design basis needs no finding of who acted or how. The class does not yet say that the holding of a structure, rather than evidence of an act, is what puts the deliberate-case footprint in force. Finding F-3.

6.6 §4.5 and §6.3 Warning and Evacuation for the Deliberate-Case Footprint

Held, with a gap. The class requires a warning that works without the dam and an evacuation plan that does not rely on a response the campaign may deny. Most of the flooded land lay under the authority that held the dam. Finding F-4.

6.7 §3.3, §6.3, §6.6 and §8.1 The Dependents of the Pool

Held, with a gap. The plant nearest to a nuclear consequence was carried by its own pond, as the class says, and then by wells drilled within about three months; water systems without a reserve of their own had none. Finding F-5.

6.8 §5.7, §6.3 and §6.7 The Pool Rule When the Dam Is Held

Held, with a gap. The pool rule names an authority to order a drawdown; the authority that could have lowered this pool was the party holding the gates. Finding F-6.

6.9 §4.4 and §4.5 The Measure of the Consequence

Held, with a gap. The footprint was measured independently of either party; the loss of life was not, and could not be. Finding F-7.

6.10 §6.7 and §6.8 The Autonomous Safe Impoundment Condition and the First Residual

Held. The first declared residual, that a sufficiently large deliberate act can breach any dam and that the response is warning, evacuation and a pool held low in advance, is what the event realized. None of the three responses was in the owner's hands.

6.11 §6.5 The Inundation Map as a Targeting Aid

Not exercised. Nothing in the sources bears on the use of inundation information.

6.12 §8.4 The Law of Armed Conflict

Held. The class records Article 56 of Additional Protocol I as setting, and does not rely on it to protect a dam; the event bears that out.

6.13 §7.2 to §7.4 The Dual Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the event, and no gate command is in the record.

6.14 Summary of Design Basis Performance

Of the thirteen readings, eight Held, with a gap, the class's account among them; three Held; and two were Not exercised. Nothing was Missed. The class anticipated the shape of the event, the two chains from one act, at a high pool, with the response denied. What it did not yet state is that the party holding a structure sets its pool, its condition and the authority over part of its valley, and that the holding, not the act, is where the design basis must begin.

7. Findings

Section references are to *Dam Structures and Water-Control Works* v0.4 unless another instrument is named.

7.1 F-1 — A Held Dam's Pool Is the Holder's, Not the Permit's

For seven months the gates were worked from the held bank at about 15 percent of the spillway's capacity, the pool was drawn to its lowest in three decades and then raised to its highest, and from about 23 April 2023 it stood above what the spillway could pass, overtopping it until the breach (Section 3.3). The finding confirms §3.3's and §4.2's choice of a high pool for the deliberate case, and is that the class should state that, for a structure held or whose gates are worked by a party to an armed conflict, the governing release is taken at the highest pool the structure and its works can physically hold, including a pool held above the spillway's capacity, and not at the highest pool the dam is permitted to hold; and that the decision margin of §3.3 is counted from the first observation that the works are no longer operated to the owner's rule.

Affected: §3.3, §4.2, §6.9. Disposition: R-1.

7.2 F-2 — Persistence Acts Through Operation, and a Denied Structure Is Still Observed

The owner could not inspect the structure for fifteen months. The spillway was overtopped for six weeks, a pier downstream of the gates was visibly damaged nine days before the breach and further structure was missing two days before it, all of it observed on commercial satellite imagery and published after the event, and no warning was issued (Section 3.4). The finding confirms the v0.4 sentence of §6.2, and is that the class should state that persistence acts through the operation of the works as well as through the denial of inspection, so that a structure held by such a party is, under a positive applicability determination, no longer credited with its shown stability from the day its operation leaves the owner's rule, without waiting on the interval §6.2 asks the design basis to state; that, while inspection is denied, the pool level and the visible condition of the spillway and its structures are followed by remote observation, satellite altimetry and imagery among it, at an interval the design basis states and by a party it names, which the boundary of §10.6 would place in OpsCo; and that observed overtopping of a spillway's structures, or observed damage below its gates, is a sign of a release for the warning time of §4.5.

Affected: §4.5, §6.2, §10.6. Disposition: R-2.

7.3 F-3 — The Design Basis Does Not Wait on Attribution

The two governments name each other; NORSAR reported an explosion at the time of the breach; Yang and colleagues found a structure eroded by six weeks of overflow; no court has ruled (Sections 2.2 and 3.4). The downstream population met the same flood on any reading, and the Emergency Action Plan's question on 6 June was the same on any reading. The finding

confirms §6.3 as the governing scenario, and is that the class should state that a high-hazard dam held by, or within reach of the works of, a party to an armed conflict carries a positive applicability determination from the day it is held, with its deliberate-case footprint, warning and pool rule in force from that day; that the deliberate breach and the progressive failure of a held structure are read together under §6.3, since the design basis cannot tell them apart in time to warn; and that no requirement of the extension waits on a finding of who acted.

Affected: §6.1, §6.3. Disposition: R-3.

7.4 F-4 — The Warning Must Reach a Valley Under Another Authority

About two-thirds of the flooded area lay in the territory held by the party that held the dam (Section 3.6). There the response was absent for days, evacuation within the town was refused, the United Nations' requests for access were declined, and the counts of the dead differ most (Sections 2.2 and 3.7); on both banks evacuations were shelled. The finding confirms §6.3's requirement for a warning that works without the dam and an evacuation that does not rely on a response the campaign may deny, and is that the class should state that, where any part of the deliberate-case footprint may lie under an authority other than the dam's owner and the warning authority, the warning plan names the paths by which a warning reaches that part without that authority's cooperation, and that the evacuation plan for that part is stated as a declared residual in the form of *Defense in Depth* §3.5 where no such path exists. This is noted to the Emergency Services series, where *Air-Raid Alerting in Ukraine* (DBA-ESS-RE-9) reads a national alerting system working across a front line.

Affected: §4.5, §6.3, §6.8; noted to *Emergency Management and Coordination* and DBA-MA-CDB. Disposition: R-4.

7.5 F-5 — A Dependent's Margin Is Its Reserve and the Time to Build Another

The reservoir fell at up to about 11 centimeters an hour toward the 12.7-meter level below which the Zaporizhzhia plant could not pump from it (Section 3.8). The plant was carried by its full cooling pond, with its reactors in shutdown, and by refilling its sprinkler ponds while the reservoir could still be reached; within about three months ten wells had been drilled to supply the sprinkler ponds. About 700,000 people lacked access to drinking water. The finding confirms §6.3's rule that the drawdown authority weighs the downstream margin against the dependents' own margins and not their stated needs, and the class's reading of Kakhovka in §6.3 and §6.6, and is that the class should state the pool-dependency margin of §3.3 as two quantities, the time the dependent's own reserve lasts and the time it needs to establish an alternative supply, with the dependent's state at the time (a reactor in shutdown, a canal in season) stated beside them; and should note to the nuclear series of DBA-MA that the margin the Zaporizhzhia plant used depended on its reactors being in shutdown, and to *Storage Reservoir* that a water system

drawing from a reservoir with no reserve of its own has a margin of zero, a finding for its own design basis (§8.1).

Affected: §3.3, §6.3, §8.1, §10.9; noted to the nuclear series of DBA-MA and *Storage Reservoir*. Disposition: R-5.

7.6 F-6 — When the Dam Is Held, the Owner's Lever Is Upstream

The pool rule of §6.3 and the drawdown of §6.7 name an authority to order the pool lowered. Through the fifteen months the gates were worked only by the party holding them (Sections 3.2 and 3.3). Yang and colleagues found that the cascade's other reservoirs stayed within their usual ranges while Kakhovka's did not; this instrument does not know what operation of the dams upstream was available or considered, and makes no finding on it. The finding confirms §6.3's requirement that the pool rule be stated in advance, and is that the class should state, for a dam in a cascade whose works may be held, the pool rule that the dams upstream still under the owner's or the regulator's control can carry, as inflow limited ahead of the high-water season, with the cost to their own dependents; and that §5.7's single plan for a cascade includes this rule.

Affected: §5.7, §6.3, §6.7. Disposition: R-6.

7.7 F-7 — The Consequence Is Validated on the Footprint, Not the Count of the Dead

The footprint was measured by the joint assessment at about 620 square kilometers, and the held share is stated by the holder itself. The count of the dead runs from 31 in the territory the Government of Ukraine controlled to 59 stated by the Russian Federation for the territory it controls to at least hundreds in one occupied town on the Associated Press's investigation, where the occupation authorities took control of the death certificates (Section 2.2). The finding confirms §4's measure of the consequence by footprint, depth and population at risk, and is that the class should state that a deliberate-case inundation model is checked against an event on the footprint, the depth and the arrival times, which can be observed independently of either party, and that a count of the dead from a deliberate event is not used to calibrate the model's estimate of loss of life, because the party that held the structure may also hold the record.

Affected: §4.4, §4.5, §10.1. Disposition: R-7.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 to R-7 dispose to *Dam Structures and Water-Control Works*; R-2's remote observation also to the framework's boundary with OpsCo; R-4 is noted to the Emergency Services series and DBA-MA-CDB; R-5 to the nuclear series of DBA-MA and *Storage Reservoir*.

No.	Title	Affected	Recommendation	Pri.

R-1	Take the held dam's release at the pool the structure can hold	§3.3, §4.2, §6.9	For a structure held or worked by a party to an armed conflict, take the governing release at the highest pool the structure and works can physically hold, above the spillway's capacity if they can, not the permitted pool; count the decision margin from the first observation that the works have left the owner's rule.	1
R-2	Count operation as persistence, and watch the denied structure	§4.5, §6.2, §10.6	State that persistence acts through operation; withdraw credit for shown stability from the day operation leaves the owner's rule; name remote observation of pool and structure, its interval and its holder (OpsCo on §10.6's boundary); make observed overtopping or damage below the gates a sign of a release.	1
R-3	Put the deliberate case in force on the holding	§6.1, §6.3	A high-hazard dam held by, or within reach of the works of, a party to an armed conflict carries a positive applicability determination from that day; read deliberate breach and progressive failure together; no requirement waits on attribution.	1
R-4	Warn the valley under another authority	§4.5, §6.3, §6.8	Where part of the footprint may lie under an authority other than the owner's and the warning authority's, name the paths that reach it without that authority, and declare the residual where none exists. Noted to the Emergency Services series and DBA-MA-CDB.	1
R-5	State the dependent's margin as reserve and rebuild time	§3.3, §6.3, §8.1, §10.9	State the pool-dependency margin as reserve time and time to establish an alternative, with the dependent's state; note to the nuclear series the shutdown condition of Zaporizhzhia's margin, and to *Storage Reservoir* the zero margin of a system with no reserve.	2
R-6	State the upstream pool rule for a held dam in a cascade	§5.7, §6.3, §6.7	For a cascade whose works may be held, state the inflow-limiting rule the dams still under the owner's or regulator's control can carry, with its cost to their dependents, inside the cascade's single plan.	2
R-7	Validate on the footprint, not the dead	§4.4, §4.5, §10.1	Check the deliberate-case model against an event's footprint, depth and arrival times; do not calibrate loss of life on counts from a deliberate event.	3

9. Open Items

OI-1: the register row. Project 0 entered the tier determination and the impacted sectors, and on 13 October 2026 recorded the event as military action without the tier; Section 5.5 now states the same. OI-2: the record is open. The war continues; the cause is contested between the two governments and is before no court whose finding this instrument has read; the count of the dead

is contested; the reservoir's future is undecided. Each bears on F-3, F-4 and F-7. OI-3: the class has not taken in this instrument and conforms at its next revision, with Erratum 1.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework. It makes no finding on who destroyed the dam or how, none of fault, and none on the counts of the dead.

10.2 Relationship to the Class's Use Case

DBA-DAMS-UC-1 (*A Deliberate Breach at High Pool with the Warning Denied*) was written before this instrument on the class's characterization of the event, and conforms to it. The denial of the warning before the act is the use case's forward construction and not a fact of this event; the sources read here establish neither a warning given nor a warning denied for the first hours of 6 June. Findings F-1 to F-4 bear on the use case's drive of §6.3, and F-6 bears on its question of how the drawdown authority weighs the downstream margin.

10.3 Relationship to the Series' Other Instruments

The flood's reach into the facilities in the footprint is the material of DBA-DAMS-UC-3 (*The Facilities in a Dam's Flood Path*). The destruction of the powerhouse is *Large Hydroelectric Storage*'s generating consequence, and its dam-safety chain is now this class's (§10.3). The Edenville and Sanford failures (DBA-DAMS-RE-3) are the cascade's case and the series' other progressive failure.

References

Government of Ukraine and United Nations, *Kakhovka Dam Destruction: Post-Disaster Needs Assessment*, released 17 October 2023, and the United Nations in Ukraine's press release of that day. International Atomic Energy Agency, Update 161 and Update 162, IAEA Director General statements on the situation in Ukraine, 6 and 7 June 2023; NucNet, report of 25 September 2023 on the Agency's account of the wells at the Zaporizhzhia plant. United Nations, statement attributable to the Secretary-General on the Kakhovka dam, 6 June 2023; United Nations Humanitarian Coordinator for Ukraine, statement of 18 June 2023, as the Times of Israel and Associated Press reported it. NORSAR, statement of June 2023 on seismic signals at the Kakhovka dam, as the Moscow Times, Arab News and NBC News reported it. Yang, Q., Shen, X., He, K., Zhang, Q., Helfrich, S., Straka, W., Kellndorfer, J. M. and Anagnostou, E. N., "Pre-failure operational anomalies of the Kakhovka Dam revealed by satellite data", *Communications Earth & Environment* 5, 230, 2 May 2024. Associated Press, investigation of 28 December 2023 into the deaths at Oleshky, as the Boston Globe reprinted it. NPR, "Russia is draining a massive Ukrainian reservoir, endangering a nuclear plant", 10 February 2023. Ukrainska Pravda, report

of 20 July 2023 of the Minister of Internal Affairs' count. Moscow Times, 9 June 2023, on the shelling of evacuations. Reports of 10 and 11 June 2023 of the Under-Secretary-General for Humanitarian Affairs' estimate of people without drinking water. Protocol Additional to the Geneva Conventions of 12 August 1949 (Protocol I), Article 56. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 13 October 2026: issued in Project 11 under the Dams Sector Framework v0.4, on the conferral of the framework's §9.1 of 12 October 2026, read against *Dam Structures and Water-Control Works* v0.4. Written new under the Reference Event Compression Standard v0.3, the series' first reference event instrument. Tier determination at Section 5.5; F-1 to F-7, R-1 to R-7. The class's account qualified at four points, which go to the class as Erratum 1 of Project 11's note of 13 October 2026 (Dams batch 1). Record open (OI-2).

Version 0.2, 13 October 2026: corrects the short title of DBA-ESS-RE-9 at F-4, to *Air-Raid Alerting in Ukraine*, found while writing the Healthcare batch. No other change; every finding, recommendation and determination is unchanged.

Version 0.3, 1 November 2026: on the author's ruling of 13 October 2026 that all four characteristics must be established, Section 5.5 is restated as military action without the tier, citing FD-DBL (*Design Basis Layers*) §6.4, and OI-1 is conformed to the register row Project 0 entered that day. The characteristics found, the attribution posture, the governing scenario, every finding and every recommendation are unchanged. Made before the instrument is frozen for the first edition of the CIKR example site. Version 0.2 is in the project's _archive.